

Intervals and absolute value equations

Hardness of the spectral-norm condition number

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Area: condition estimation and rigorous error bounds

Rating rationale: Challenging reflects an exact complexity classification under a regularity promise where other norms behave differently; specialist impact is the computation of one AVE condition number.

Independently reviewed resolution - 2026-09-11

Hardness classification. Theorem 2 and Sections 2-3 give a polynomial-time many-one reduction from MAX-CUT to the threshold $c_2(A) \geq t$, using integer upper-triangular matrices with diagonal 2. All queried families are regular. This proves the requested promise-preserving Turing hardness, with equality in the yes case; the restricted rational triangular problem is NP-complete. The result does not assert $P \neq NP$.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. See the [complete manuscript](#), [independent agent review](#) and [submission record](#). The source archive identifies the drafts as AI-generated; authorship is recorded at the submitter's request. This is agent verification, not external human peer review or formal proof-assistant certification.

The difficulty, importance and rating rationale below are historical assessments of the original open target. The original statement and dated audits are preserved.

Context and notation

Absolute values and vector inequalities are componentwise. All algorithmic questions use rational input in binary and the deterministic Turing model. Input length includes the matrix dimensions and the bit lengths of all rational coefficients.

For a real square matrix A , define the diagonal perturbation family

$$\mathcal{D}(A) = \{A - \text{diag}(d) : d \in [-1, 1]^n\}.$$

Call this family **regular** when every member is nonsingular. Only diagonal entries vary. This is precisely the entrywise interval matrix $[A - I_n, A + I_n]$.

Problem statement

Let $n \geq 1$ and let $A \in \mathbb{Q}^{n \times n}$ be promised to have regular $\mathcal{D}(A)$. Define

$$c_2(A) = \max_{d \in [-1, 1]^n} \|(A - \text{diag}(d))^{-1}\|_2 = \left(\min_{d \in [-1, 1]^n} \sigma_{\min}(A - \text{diag}(d)) \right)^{-1}.$$

Question

Is deciding $c_2(A) \geq t$, given such an A and a positive rational threshold t , NP-hard under polynomial-time Turing reductions that query only inputs satisfying the regularity promise?

This is a decision formulation of the published hardness conjecture. Equality belongs to the yes case. Checking the promise is outside the task. The spectral norm is the operator norm induced by the Euclidean vector norm.

References

Moslem Zamani and Milan Hladík, *Error bounds and a condition number for the absolute value equations*, *Mathematical Programming* **198** (2023), 85–113, §2, paragraph before Proposition 3; §2.1 and §7. The maximum is attained at a sign vector by Proposition 2; Proposition 6 gives a formula for symmetric A . The conjecture concerns general matrices.

The quantity bounds forward error by residual for $Ax - |x| = b$; see Theorem 7. Its computation remains explicitly unclassified in Hladík et al.’s [2026 survey](#), §5.4.

Earlier status evidence — 2026-09-08

The original paper’s latest arXiv version is [v2, January 17, 2020](#); the journal text was published January 30, 2022. Searches for “absolute value equations condition number 2026”, “spectral norm complexity”, and “2-norm NP-hard” found no resolution.

Audit update — 2026-09-10

Rechecked the [2025 survey](#), §5.4, and the [condition-number paper](#). They distinguish the unresolved spectral norm from proved hardness for the one- and infinity-norm versions. Spectral-norm complexity searches found no resolution under the stated promise. Impact is narrowed to specialist to match this particular condition-estimation classification.